Final Draft of Report

Project #2: Descriptive Analysis of St. Louis Area Restaurants

Joshua VanNostrand & Idirissa Diallo
MGIS 355: Business Intelligence | Professor Bui

Introduction

In this report, we focused our analysis on restaurant businesses in the St. Louis area and to better understand the market we used specific data and customer reviews. First, we used cluster analysis to group restaurants based on significant numerical factors such as the restaurant ratings, the number of reviews, pricing, and service features. With this we were able to see the similarity and differences between restaurants in the area. The second and final analysis was a text analysis done on customer reviews, in which we examined the entirety of the sentiment and found out what topics customers talked about. With both of these unique approaches, we are able to understand the structure of these restaurants, what customers think and value in their experiences. Overall, it provided us with a more complete picture of the restaurant landscape in St. Louis.

Section 1: Cluster Analysis of Restaurants in St. Louis

In this section we wanted to analyze certain patterns among restaurants in the St. Louis area, so we performed a cluster analysis using data on businesses from a website called Yelp. Specifically, we focused only on businesses categorized as restaurants. We decided to target restaurants only because that way we are able to ensure that all observations share similar characteristics, such as pricing strategy, customer reviews, and service related features and this makes the clusters more accurate and comparable. Restaurants also provided the most complete and consistent set of numeric and attribute data within the dataset which made conducting a cluster analysis reasonable.

For the clustering process, we selected key numeric variables that capture both customer perception and business scale, including review count (as a measure of popularity and customer engagement), rating (as a measure of customer satisfaction), and price level (as an indicator of market positioning). In addition to these variables, we incorporated business attributes by converting them from categorical values into numeric form. The relevant features that were chosen such as delivery availability, outdoor seating, reservations, and other service options were transformed into binary indicators and then aggregated into a single feature count variable for each restaurant. This allowed us to include service related characteristics in a common space for clustering.

Before applying the clustering algorithm, the selected variables were scaled to ensure that differences in size (for example, between review counts and ratings) did not disproportionately

influence the results. We then applied the K-Means clustering algorithm, which groups restaurants by minimizing the distance between observations within the same cluster while maximizing separation between clusters. To determine the appropriate number of clusters, we used the Elbow Method, evaluating how the within cluster sum of squares changes as the number of clusters increases. Based on this analysis, three clusters were selected as they provided a clear balance between keeping and the model simple and accurately differentiated across restaurant groups.

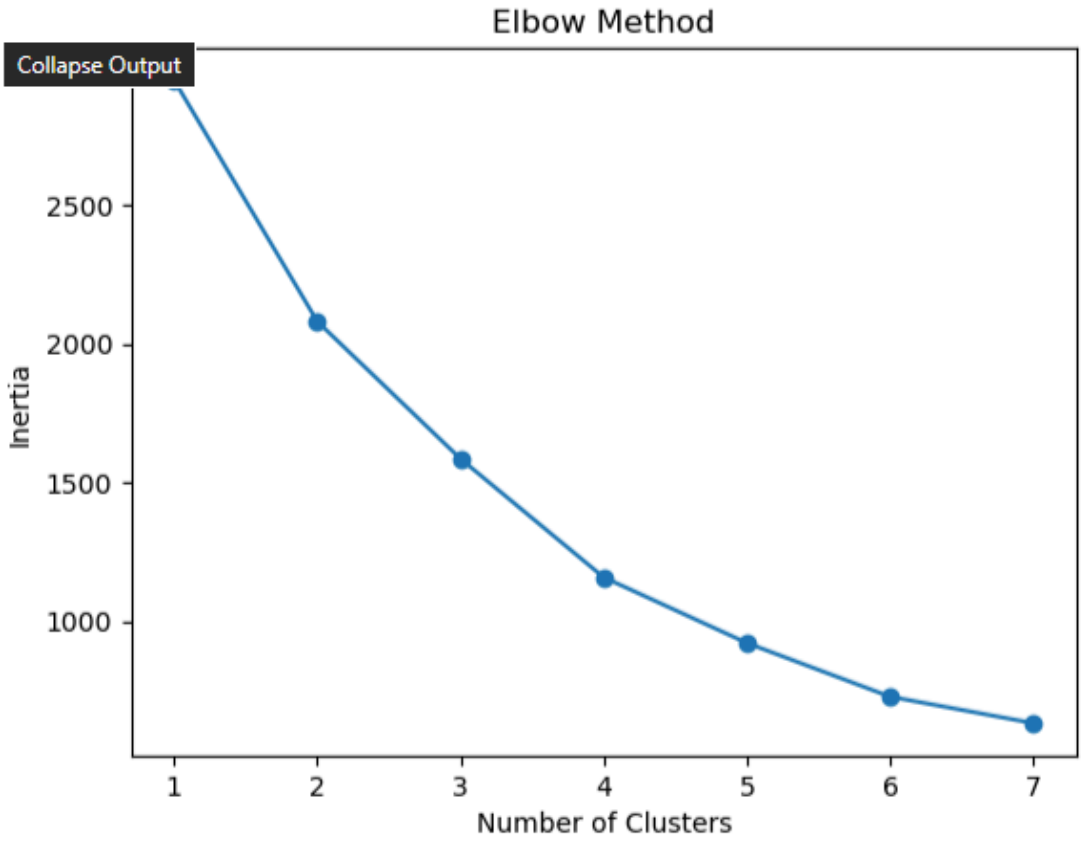


Figure 1: Elbow Method - Number of Clusters vs. Inertia

Cluster Profile and Interpretation

The cluster analysis resulted in three clear groups of restaurants in St. Louis.

Cluster 1: Popular & Premium includes higher end restaurants with high ratings, medium to high prices, and a large number of reviews. These restaurants are usually well established and attract loyal customers who are looking for a higher quality dining experience.

Cluster 2: Casual & Mid Range represents everyday restaurants with average ratings, relatively cheaper prices, and a decent number of reviews. These restaurants serve customers who want affordable and convenient dining options.

Cluster 3: Niche & Low Visibility includes smaller restaurants with less reviews, lower ratings, and cheaper prices. These are often new, a bit more specialized, or less well known restaurants by the people.

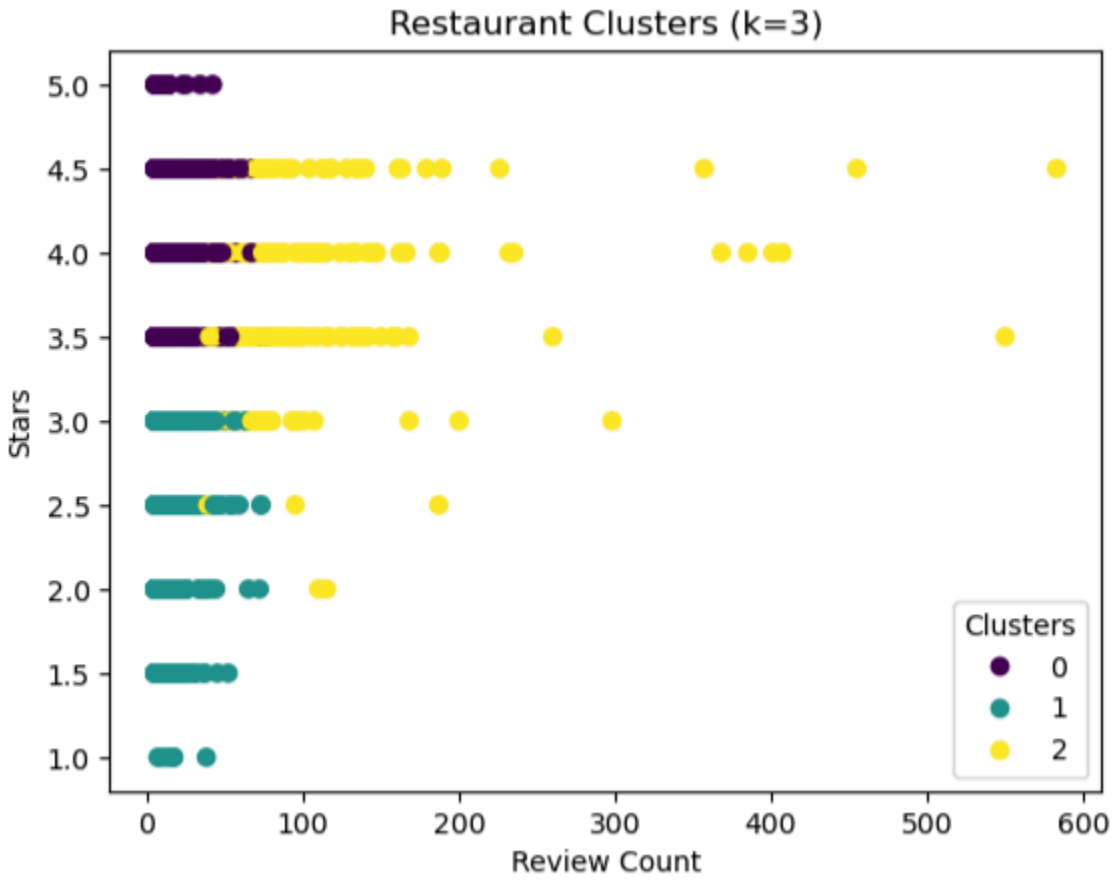


Figure 2: Restaurant Clusters: Starts vs. Review Count



Figure 3: Cluster Membership Distribution

The clusters show some clear patterns. Restaurants in Cluster 1 are more visible and get a lot of positive reviews, which means they have a strong reputation. Cluster 2 has average visibility and reviews, which fits what we expect from casual, mid range restaurants. Cluster 3 includes restaurants that are less visible and have fewer reviews, which suggests they may have room to grow. We also see that features like delivery, outdoor seating, and reservations are more common in Clusters 1 and 2, showing that restaurants with more features tend to be more popular.

cluster	attributes.RestaurantsPriceRange2	attributes.RestaurantsTakeOut	attributes.RestaurantsDelivery	attributes.OutdoorSeating	attributes.WiFi	attributes.RestaurantsReservations	attributes.Alcohol	attributes.GoodForKids	attributes.RestaurantsGoodForGroups
0	1.425743	0.845570	0.296203	0.377215	0.0	0.215190	0.339241	0.650633	0.640506
1	1.350140	0.846734	0.660804	0.251256	0.0	0.115578	0.233668	0.753769	0.685930
2	1.824468	0.973684	0.436842	0.636842	0.0	0.410526	0.810526	0.857895	0.963158

Figure 4: Cluster Attribute Profiles Table

Summary

Our cluster analysis of St. Louis restaurants shows some clear patterns in customer activity, pricing, and service features. The three groups, Popular & Premium, Casual & Mid Range, and Niche & Low Visibility highlight differences in how well known restaurants are, their ratings, and whether they offer features like delivery or reservations. Restaurants with higher ratings and more reviews tend to be more visible and seen as higher quality, while smaller or newer restaurants usually have fewer reviews and are less visible. Overall, this analysis helps us better understand the restaurant market by showing where competition is strong and where there may be room to grow. When we combine these results with other analyses, like review sentiment and topic modeling, we get a more complete picture of how restaurants perform and what matters most to customers in St. Louis.

Section 2: Text Analysis of Restaurant Reviews

To complement the cluster analysis, a text based analysis was conducted on Yelp reviews for restaurants in the St. Louis, Illinois area. Two methods were applied: sentiment analysis using VADER and topic modeling using LDA. The decision to focus exclusively on restaurant reviews was deliberate. Restaurants represent the largest and most consistently reviewed business category in the dataset, making them the most data rich segment for text analysis. Analyzing a single category also ensures that the insights drawn are directly comparable across businesses rather than being diluted by reviews from unrelated industries such as retail or healthcare.

VADER was selected for sentiment analysis because it is specifically designed for social media and review style text, making it well suited for informal, conversational writing like Yelp reviews. Unlike traditional sentiment tools, VADER handles slang, punctuation emphasis, and mixed sentiment effectively without requiring labeled training data. LDA was chosen for topic modeling because it is an unsupervised method that can identify latent themes across a large volume of text without prior assumptions about what those themes are. Together, these two methods provide both an emotional and a thematic lens through which to understand what customers are actually saying about St. Louis restaurants.

Sentiment Analysis Results

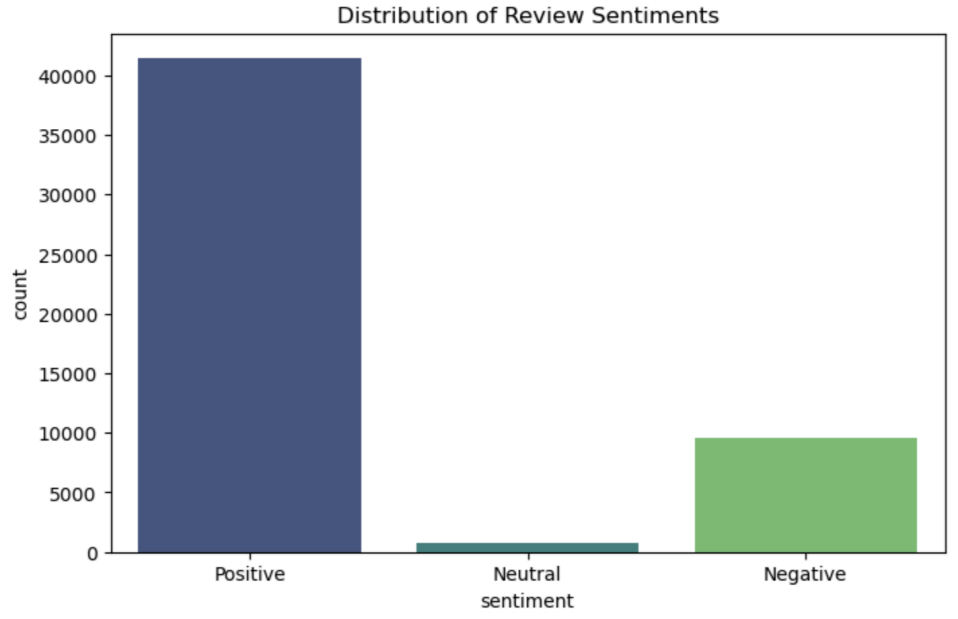


Figure 5: Distribution of Review Sentiments

The sentiment analysis revealed that the overwhelming majority of restaurant reviews in the St. Louis area are positive. As shown in Figure 5, approximately 41,500 reviews were classified as positive, roughly 9,700 were classified as negative, and fewer than 1,000 were classified as

neutral. This means that around 80% of all restaurant reviews carry a positive sentiment, which is an encouraging signal for the overall health of the restaurant market in this area.

The high proportion of positive reviews suggests that St. Louis restaurants are generally meeting or exceeding customer expectations at a broad level. However, the nearly 9,700 negative reviews represent a meaningful volume of dissatisfied customers that should not be overlooked. These negative reviews are where the most actionable business intelligence tends to live, as customers who take the time to leave a negative review are usually pointing to a specific, fixable problem. The relatively small neutral category indicates that most customers who leave reviews feel strongly enough one way or the other to express a clear sentiment, which reflects the emotionally engaged nature of dining experiences.

Topic Modeling Results

To better understand what specific aspects of the restaurant experience customers are talking about, LDA topic modeling was applied to the cleaned review text. After tuning the model across a range of topic counts and evaluating coherence scores, five topics were identified as the optimal number for this dataset. Each topic was interpreted based on its highest weighted keywords.

The five topics and their interpretations are as follows:

- **Topic 0 (food, good, great):** General food quality and overall satisfaction
- **Topic 1 (time, back, told):** Service experience, wait times, and return visits
- **Topic 2 (great, staff, always):** Staff quality and consistency of service
- **Topic 3 (food, table, u):** Dine-in experience and table service
- **Topic 4 (like, ordered, good):** Ordering experience and food expectations

These five topics map onto the core dimensions of the restaurant experience: food quality, service speed, staff behavior, dine in atmosphere, and the ordering process. The fact that the model naturally surfaced these categories without any prior labeling validates that these are the areas customers most frequently and consistently write about.

Sentiment Distribution by Topic

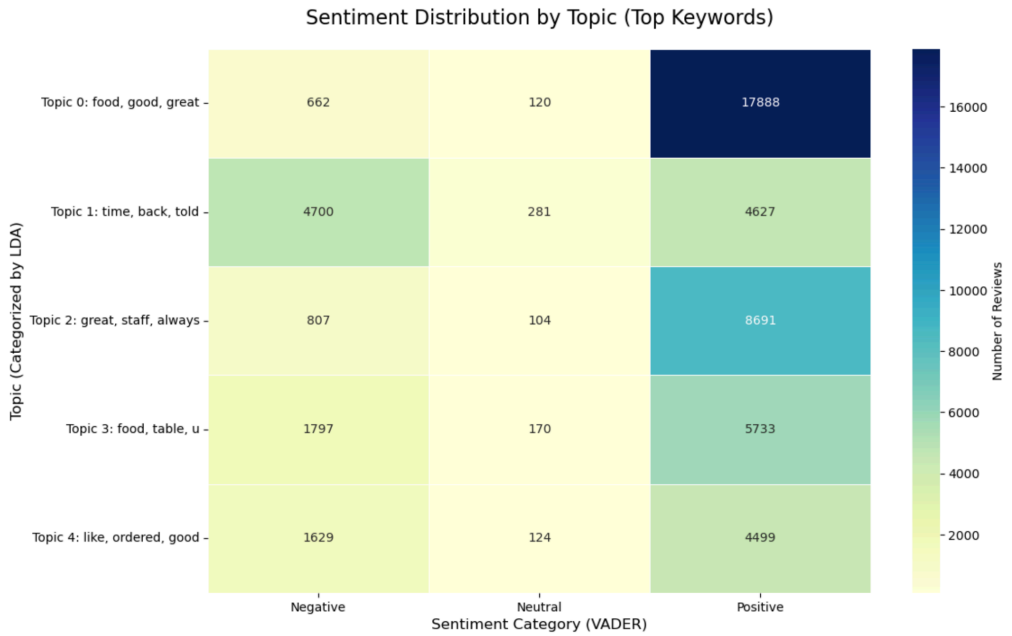


Figure 6: Sentiment Distribution by Topic (Top Keywords)

Figure 6 breaks down the sentiment distribution across each of the five topics, revealing important differences in how customers feel about each area of the restaurant experience. Topic 0, centered on general food quality, shows by far the highest volume of positive reviews at 17,888, more than double any other topic. This indicates that food quality is both the most discussed and most positively received aspect of St. Louis restaurants, which is expected but still useful to confirm.

Topic 1, which captures service experience and wait times, tells a very different story. It is the only topic where negative reviews (4,700) nearly match positive reviews (4,627), making it essentially split down the middle. This is a significant finding, it suggests that while customers are generally happy with the food, service experience is a major source of frustration and a clear area where restaurants are falling short. Topics 3 and 4, covering dine in experience and the ordering process respectively, also show elevated negative counts relative to other topics, further reinforcing that the operational and service side of restaurants is where customer dissatisfaction is concentrated.

Topic 2, focused on staff quality, leans more positive with 8,691 positive reviews against 807 negative, suggesting that when customers specifically comment on staff, they are more often praising them than criticizing them. This distinction between staff quality and overall service experience is worth noting, customers may appreciate individual staff members while still being frustrated with wait times, order accuracy, or overall flow.

Business Recommendations

Based on the combined findings from the sentiment analysis and topic modeling, the following recommendations are directed at restaurant owners and operators in the St. Louis area:

Address service speed and wait time issues. Topic 1 is the clearest problem area in the entire analysis. The near-even split between positive and negative sentiment around time, being told things, and return visits points directly to wait time management and communication with customers during their experience. Restaurants should evaluate their front-of-house operations, staffing during peak hours, and how staff communicate delays to tables. Small improvements here could meaningfully shift sentiment in the most contested topic area.

Protect and promote food quality as a competitive strength. Topic 0 is the strongest performing area by a wide margin. Food quality is clearly what St. Louis restaurant customers care about most and where they are most satisfied. Restaurants should continue investing in menu quality and consistency, and should lean into this strength in their marketing. Positive word of mouth around food appears to be driving the overall positive sentiment baseline across the market.

Use staff performance as a differentiator. Topic 2 shows that customers notice and appreciate great staff, and positive staff reviews are more common than negative ones. Restaurants that invest in staff training, consistency, and culture have an opportunity to build a strong reputation around service that complements their food quality. Given how competitive the restaurant market is, staff experience is one of the few areas that is genuinely difficult for competitors to replicate quickly.

Improve the dine in and ordering experience. Topics 3 and 4 both show meaningful negative sentiment around the physical dining and ordering experience. This could reflect issues with table turnover, order accuracy, menu clarity, or the overall pace of service during a meal. Restaurants should solicit more specific feedback from dine in customers and look for patterns in complaints around these touchpoints.

Summary

The text analysis of St. Louis restaurant reviews reveals a market that is performing well on the fundamentals, food quality is strong and most customers leave satisfied, but faces consistent challenges in the operational and service dimensions of the dining experience. Sentiment analysis confirmed that roughly 80% of reviews are positive, while topic modeling identified service experience as the single most contested area of customer feedback. Together, these findings point toward a clear opportunity: St. Louis restaurants that can match their food quality with an equally reliable and efficient service experience are well positioned to stand out in a competitive market and convert occasional visitors into loyal repeat customers.